Computer Systems - Quiz 3 (Graded)

DEMO DATA - fictional student and course, generated for the MentorIA demo.

Q1.
Topic: Caches & Locality
Question: Which loop order over a row-major 2D array has better spatial locality?
Student answer: Row-major (inner loop over columns)
Result: Correct
Score: 2/2
Grader comment: Good.

Q2.
Topic: Cache Coherence
Question: Under MESI, a core holds line X in Shared. Another core issues BusRdX for X. What is the new state?
Student answer: Stays Shared
Result: Incorrect
Score: 0/2
Grader comment: BusRdX invalidates other copies: S -> I.

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Q3.
Topic: Cache Coherence
Question: Why does a write to a line in Shared state require a bus transaction?
Student answer: It doesn't if the line is clean
Result: Incorrect
Score: 0/2
Grader comment: Other sharers must be invalidated (BusUpgr/BusRdX).

Q4.
Topic: Virtual Memory
Question: What does the TLB cache?
Student answer: Recent virtual-to-physical page translations
Result: Correct
Score: 2/2

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Q5.
Topic: Synchronization
Question: Two threads run counter++ 1000 times each without a lock. Can the final value be < 2000?
Student answer: No, ++ is atomic on x86
Result: Incorrect
Score: 0/2
Grader comment: Load/add/store can interleave: a race condition.

Q6.
Topic: Virtual Memory
Question: What happens on a page fault for a valid but non-resident page?
Student answer: OS loads the page from disk and restarts the instruction
Result: Correct
Score: 2/2

Q7.
Topic: Processes & Signals
Question: What does fork() return in the child process?
Student answer: The parent's PID
Score: 1/2
Grader comment: Partial credit: returns 0 in the child.